



Recurrent Neural Networks

Sequential Data Modeling

- ▶ **Sequential Data**

- ▶ Most of data are sequential
- ▶ Speech, Text, Image, ...

- ▶ **Deep Learnings for Sequential Data**

- ▶ **Convolutional Neural Networks (CNN)**
 - ▶ Try to find local features from a sequence
- ▶ **Recurrent Neural Networks: LSTM, GRU**
 - ▶ Try to capture the feature of the past

Sequential Data Processing

▶ What is sequential data?

1 2 3 4

Sequential data

2 ϵ
→ |

Non-sequential data ??

Sequential Data Processing

- ▶ Are both the same data?

1 2 3 4

4 3 2 1

- ▶ Same: both datasets have the same values
- ▶ Not Same: values are same,
but the positions of each value are different

Sequential Data Processing

- ▶ What do we have to consider for sequential data processing?

(x_1, y_1)

(x_2, y_2)

(x_3, y_3)

(x_4, y_4)

(x_5, y_5)

Sequential data

(x_1, y_1)

(x_2, y_2)

(x_3, y_3)

(x_4, y_4)

(x_5, y_5)

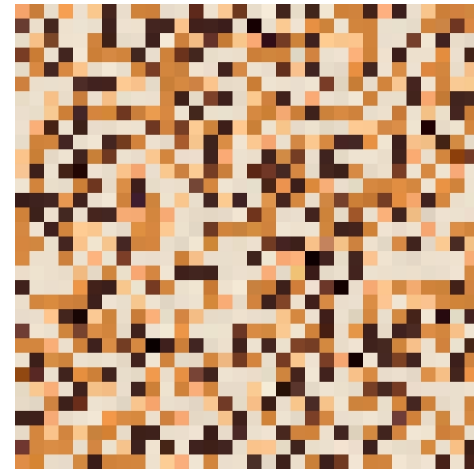
Non-sequential data

Sequential Data Processing

- ▶ What do we have to consider for sequential data processing?



Cat



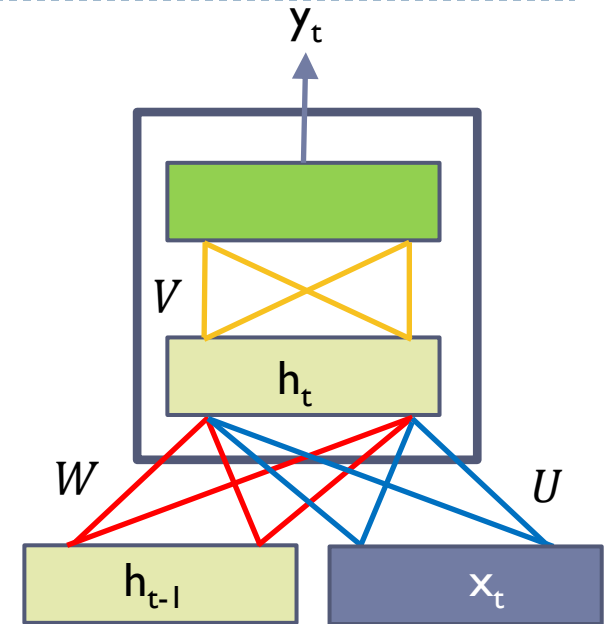
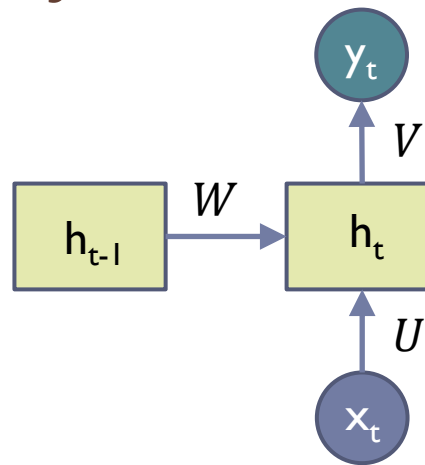
Pixel Shuffled Cat

Recurrent Neural Networks

► Connections form cycles

$$h_t = f(Ux_t + Wh_{t-1})$$

$$y_t = g(Vh_t)$$



- x_t : input at time t
- h_t : hidden state at time t
- f : is an activation function
- U, V, W : network parameters
 - RNN shares the same parameters across all time steps
- g : activation function for the output layer

Recurrent Neural Networks

► Connections form cycles

Training Data

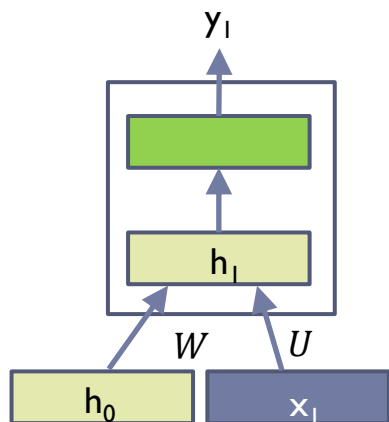
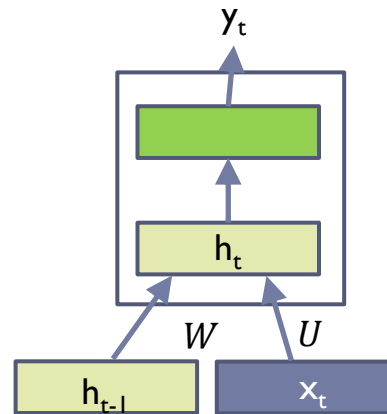
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(x_3, y_3)

(x_4, y_4)

(x_5, y_5)



Recurrent Neural Networks

► Connections form cycles

Training Data

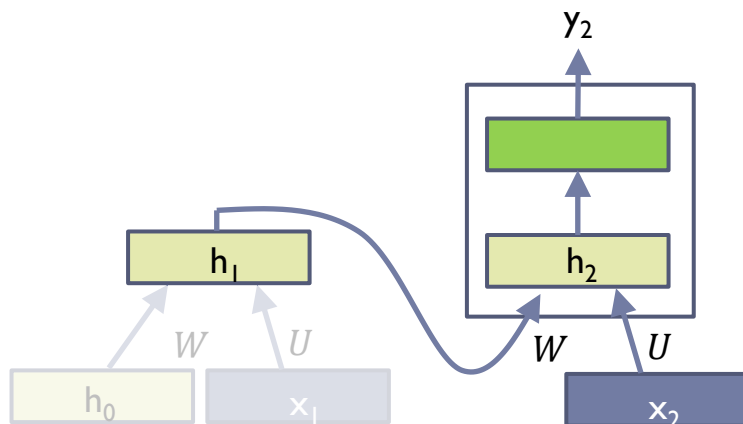
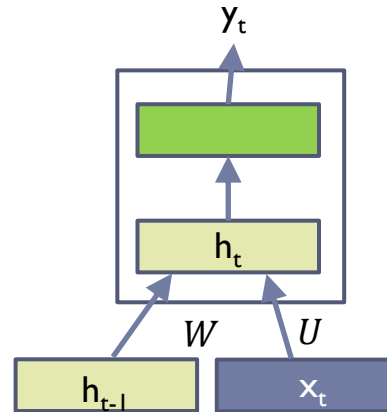
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(x_3, y_3)

(x_4, y_4)

(x_5, y_5)



Recurrent Neural Networks

► Connections form cycles

Training Data

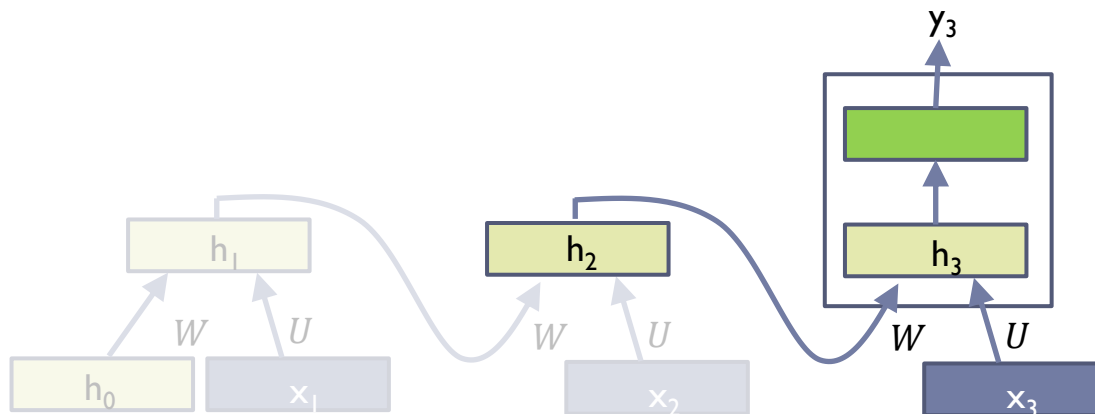
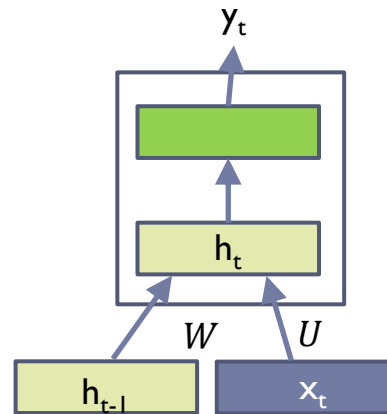
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Recurrent Neural Networks

► Connections form cycles

Training Data

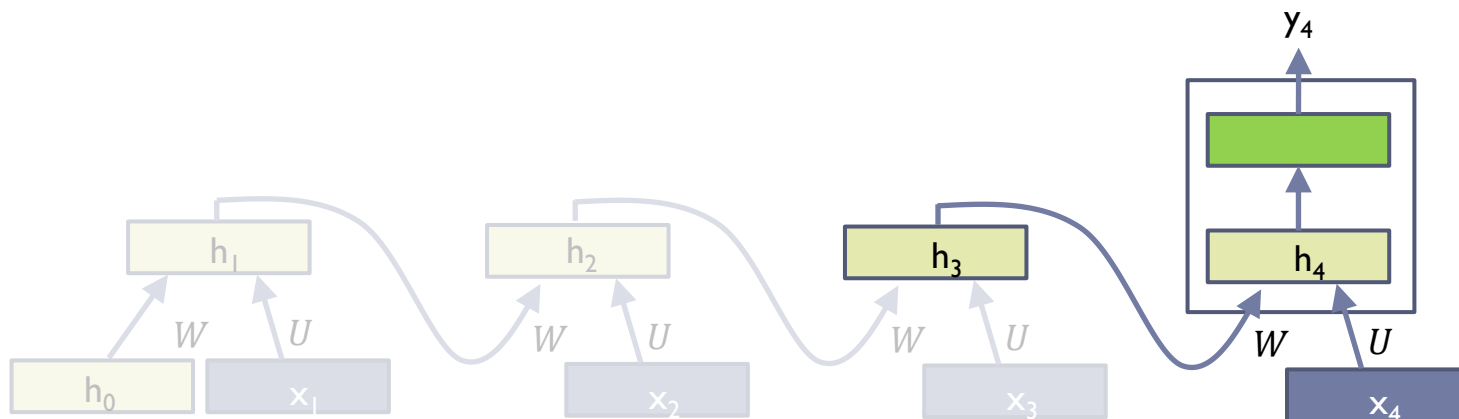
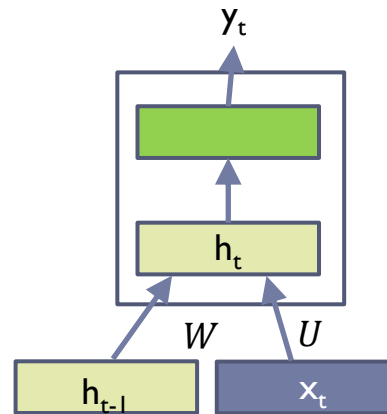
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(x_2, y_2)

(x_3, y_3)

(x_4, y_4)

(x_5, y_5)



Recurrent Neural Networks

► Connections form cycles

Training Data

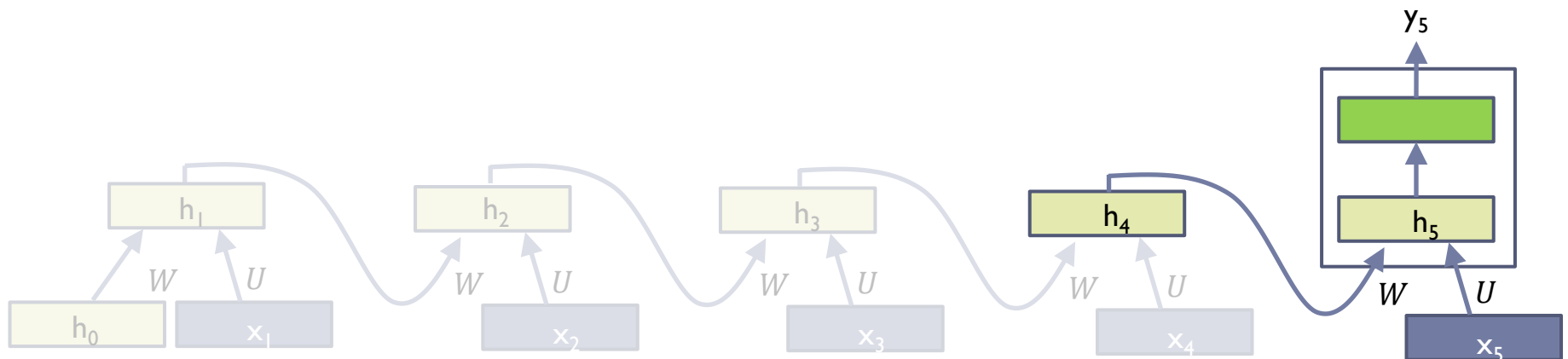
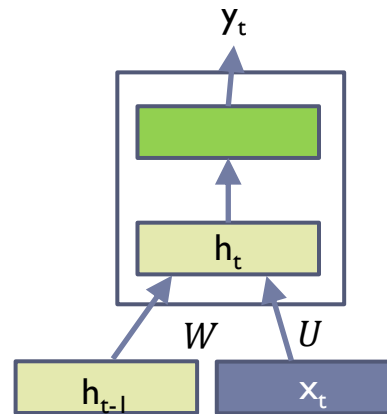
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Recurrent Neural Networks

► Connections form cycles

Training Data

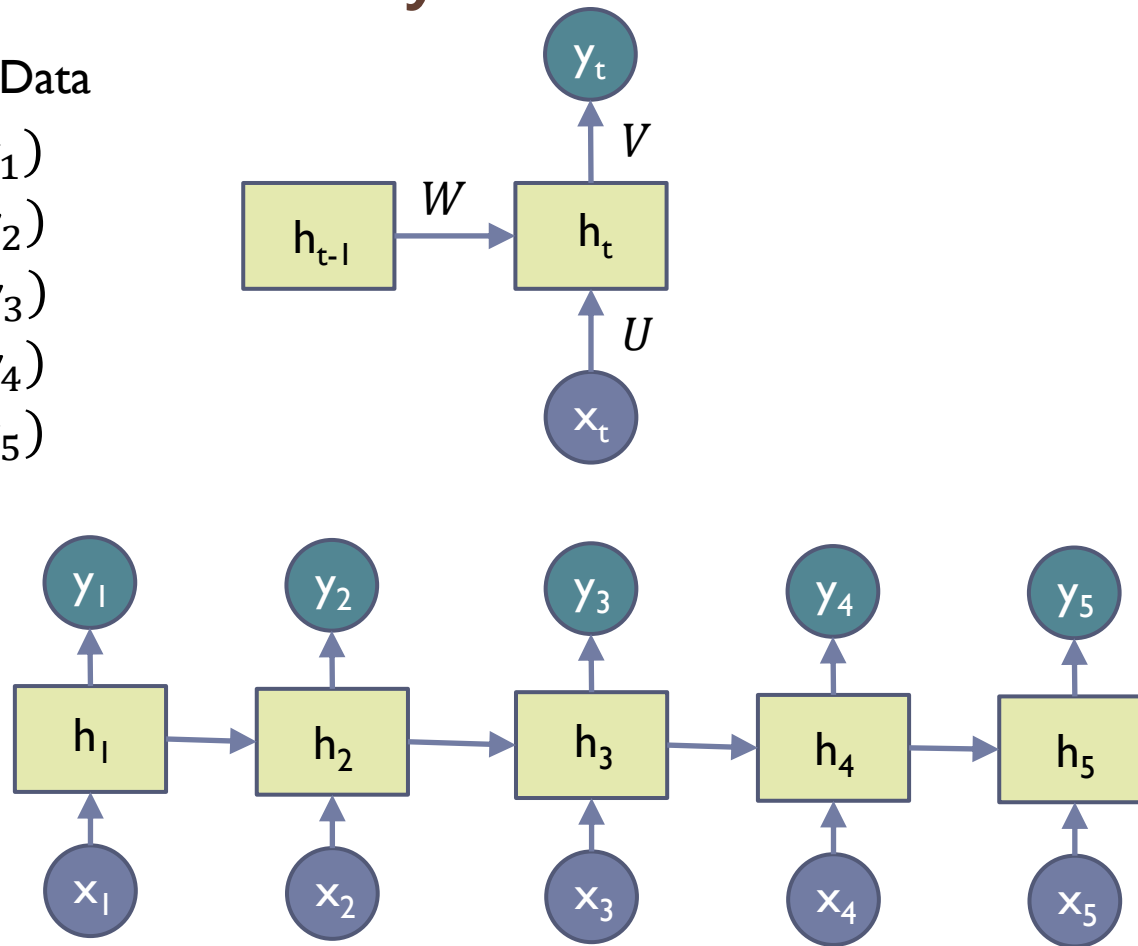
(x_1, y_1)

(x_2, y_2)

(x_3, y_3)

(x_4, y_4)

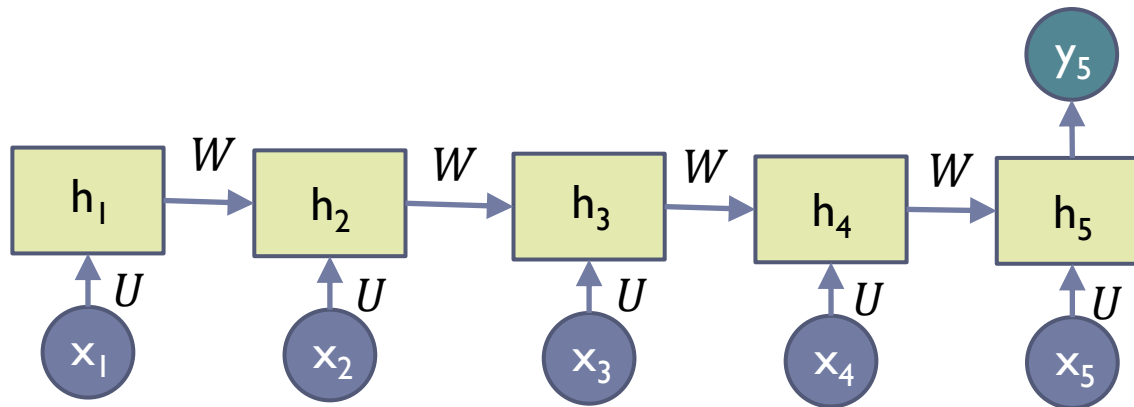
(x_5, y_5)



Recurrent Neural Networks

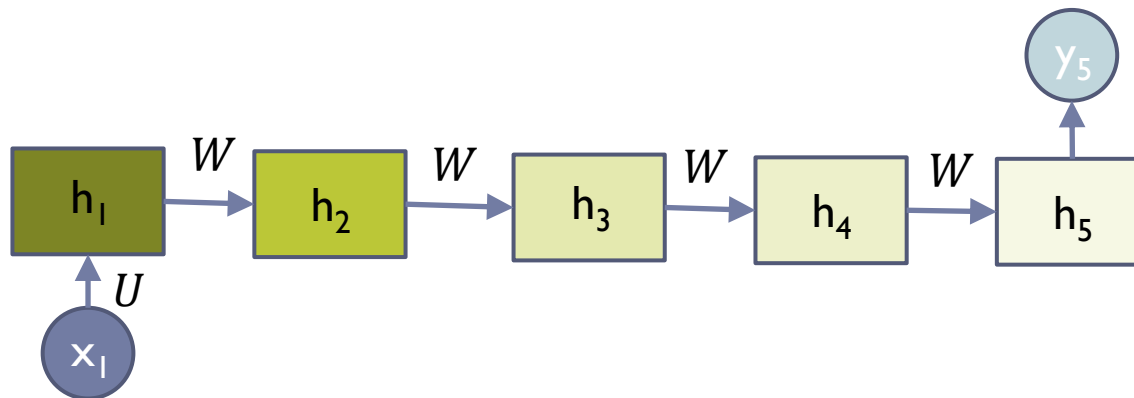
▶ Long Term Dependency

- ▶ $x_1 \sim x_{t-1}$ are encoded into h_{t-1}
- ▶ h_{t-1} has the information on the past
- ▶ It is a context to process x_t



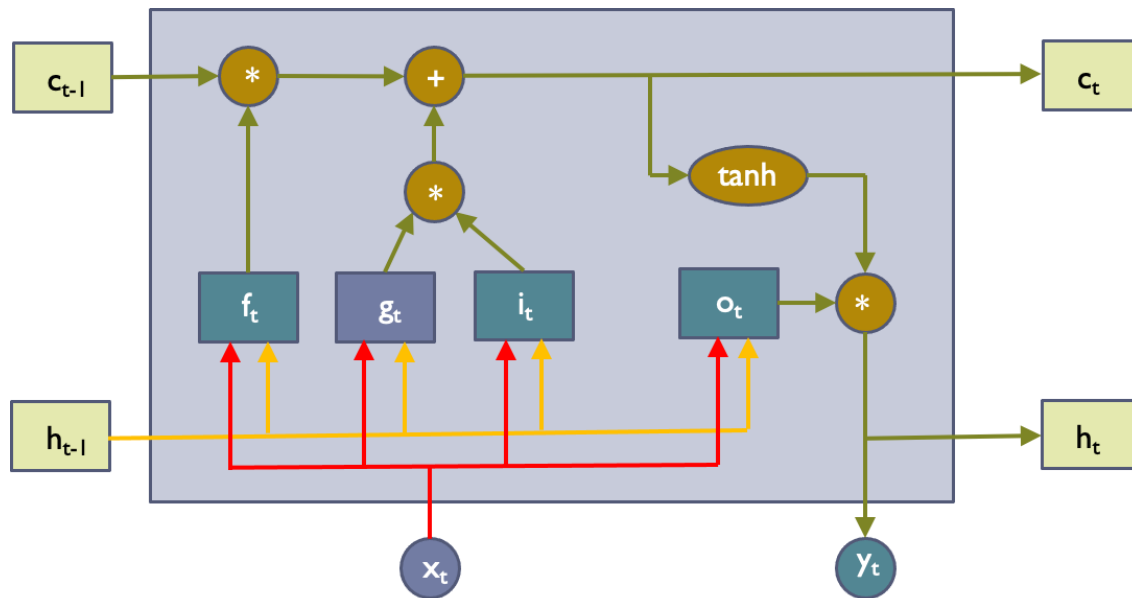
Recurrent Neural Networks

- ▶ **Long Term Dependency of Standard RNN**
 - ▶ However, it may exponentially decay or grow
 - ▶ Usually, it is limited to 10 steps



Long Short-Term Memory (LSTM)

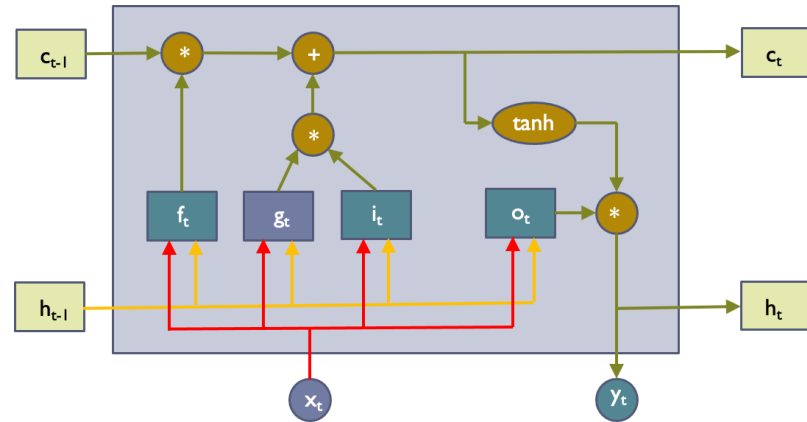
- ▶ Capable of learning long-term dependencies.
 - ▶ An LSTM can learn to bridge time intervals of 1000 steps
 - ▶ Gate units that learn to open and close access to the past



Long Short-Term Memory (LSTM)

► Equations

- i : input gate
- f : forget gate
- o : output gate
- g : self-recurrent
- c_t : internal memory
- h_t : hidden state
- y : final output

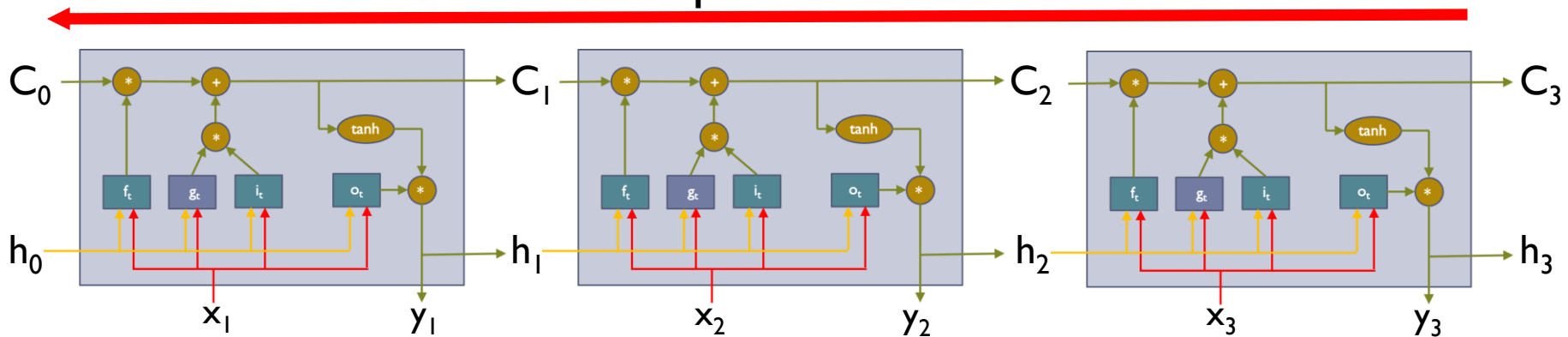


$$\begin{aligned}i &= \sigma(x_t U^i + h_{t-1} W^i) \\f &= \sigma(x_t U^f + h_{t-1} W^f) \\o &= \sigma(x_t U^o + h_{t-1} W^o) \\g &= \tanh(x_t U^g + h_{t-1} W^g) \\c_t &= c_{t-1} \circ f + g \circ i \\h_t &= \tanh(c_t) \circ o \\y &= \text{softmax}(V h_t)\end{aligned}$$

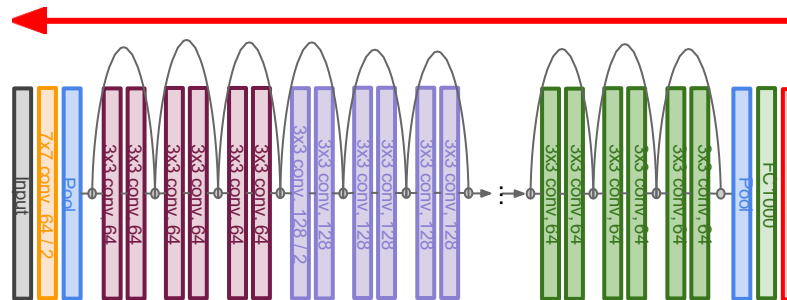
Long Short-Term Memory (LSTM)

► Gradient Flow

Uninterrupted Gradient Flow



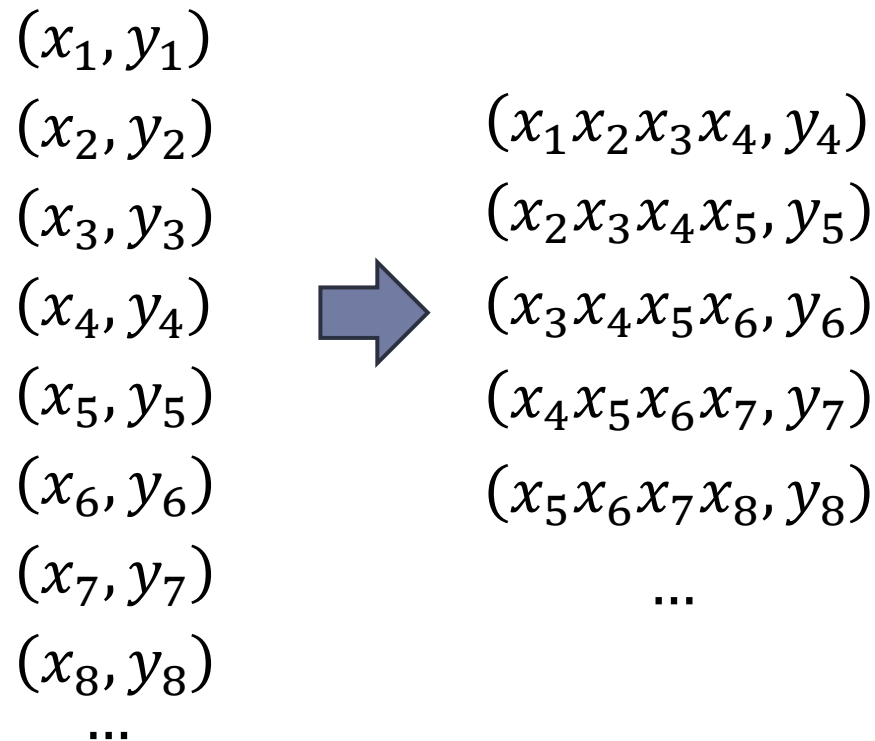
Similar to ResNet!



Sequence Processing

▶ Training Data

- ▶ Usually, samples are preprocessed in a fixed length



Sequence Processing

▶ Training

- ▶ Samples are trained with a fixed length of RNN

$(x_1 x_2 x_3 x_4, y_4)$

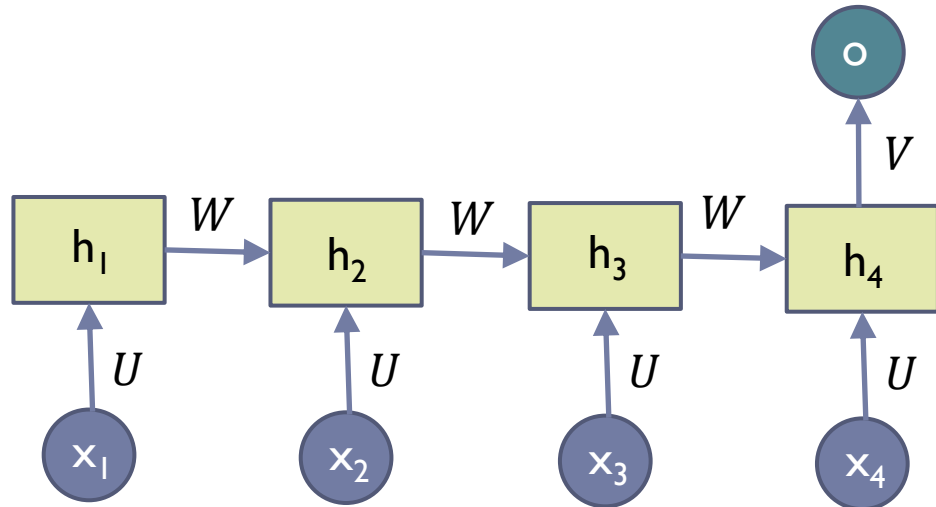
$(x_2 x_3 x_4 x_5, y_5)$

$(x_3 x_4 x_5 x_6, y_6)$

$(x_4 x_5 x_6 x_7, y_7)$

$(x_5 x_6 x_7 x_8, y_8)$

...



$$E = (y - o)^2$$

Question and Answer